Exam Date & Time: 30-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

VLSI TESTING AND TESTABILITY [ECE 3125]

Marks: 50 **Duration: 180 mins.**

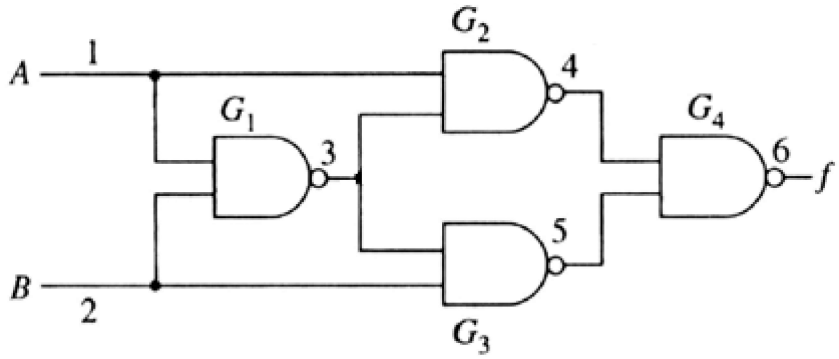
A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) For the combinational circuit shown in Fig. 1A with inputs A and B and output f, where node 3 has a stuck-at-0 fault, use the D-Algorithm to determine the test vector that activates the fault and propagates it to the output.

A)



(4)

Fig. 1A

B) Define a bridging fault and illustrate its impact on circuit behavior. Provide an example to demonstrate how the fault affects the overall circuit operation. (3)

C) Analyze the circuit in Fig. 1C to determine the fault collapse ratio using fault equivalence. (3)

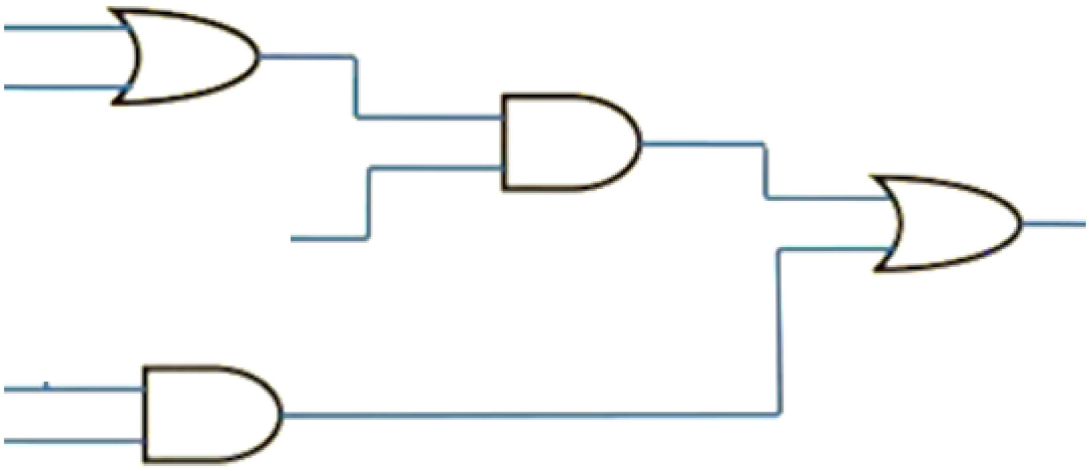
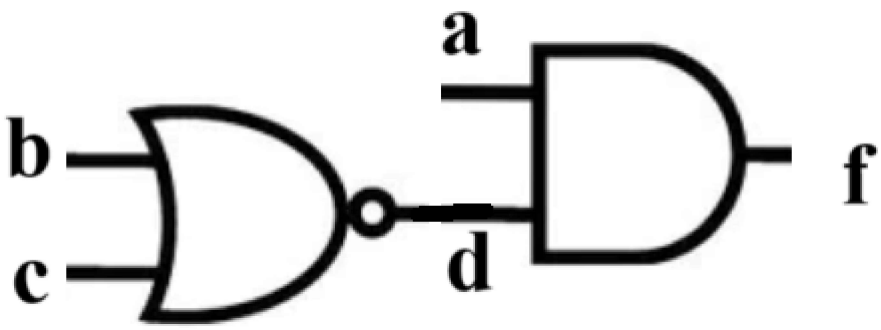


Fig. 1C

2) Construct the fault table for the circuit shown in Fig. 2A, considering stuck-at-0 (SA0) and stuck-at-1 (SA1) faults on nodes a, b, c, d, and f. Determine the minimum set of test vectors required to detect all identified faults.

A)



(4)

Fig. 2A

B) Design the scan path circuit for a sequence detector that utilizes two D flip-flops and necessary logic gates, where D1 = y2x' + y1y2'x and D2 = x, and the output Z = x'y1y2. Implement the scan path to test the stuck-at-1 fault on D1. The state transition of fault free circuit is given in Table 2B.

Table: 2B

PS	NS, z	
	x = 0	x = 1
A	A, 0	B, 0
B	C, 0	B, 0
C	A, 0	D, 0
D	C, 1	B, 0

(3)

C) Explain the concept of a critical path. For the circuit shown in Fig. 2C, identify the critical path and calculate its delay, considering the propagation delays indicated inside each logic gate symbol (3)

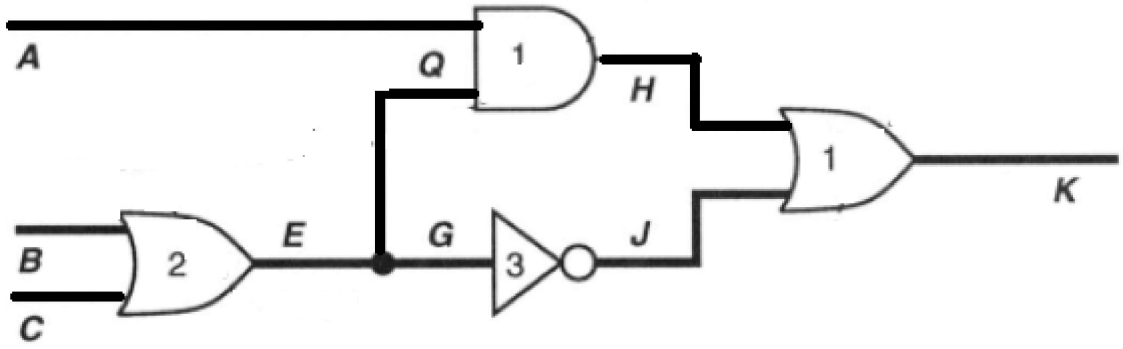


Fig. 2C.

3) For the circuit depicted in Fig. 3A, with initial contents of LFSR1 = 100 and LFSR2 = 000, determine the signature of both the fault-free and faulty circuit. (4)

A)

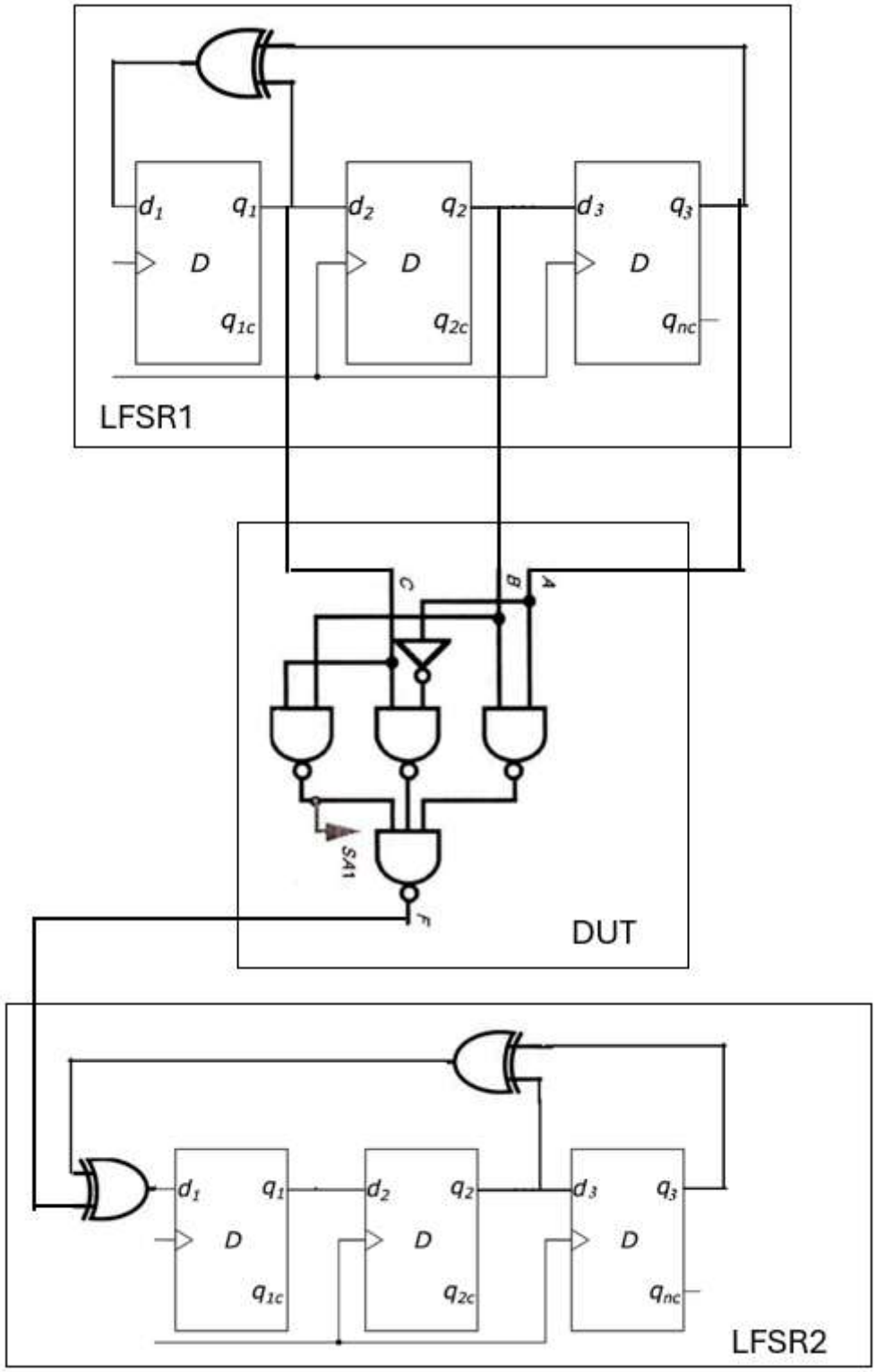


Fig. 3A.

B) Calculate the maximum operating frequency for the circuit shown in Fig. 3B, where each gate has a delay of 10 ns. Use given input transition for the calculation. (3)

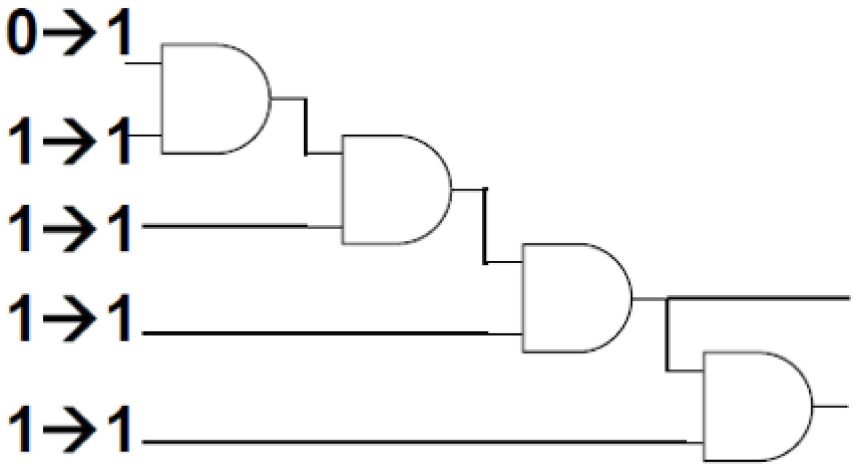
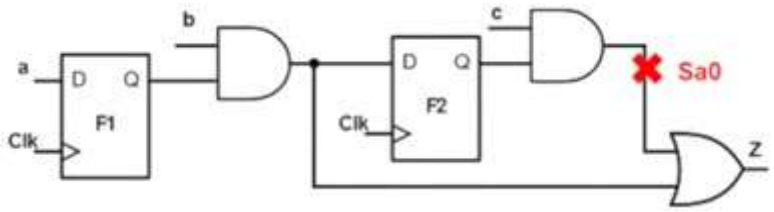


Fig. 3B.

C) Write the difference between permanent and non-permanent fault. (3)

4) Find the test vector for the Stuck-At-0 (SA0) fault shown in Fig. 4A.

A) 

(4)

Fig. 4A.

B) Construct the LFSR corresponding to the polynomial $1 + x^2 + x^4$, assuming the initial content is 0001. List the sequence of random patterns generated by this LFSR. (3)

C) For the circuit shown in Fig. 4C, L is stuck-at-1 (SA1) determine the test vector using the FAN algorithm (3)

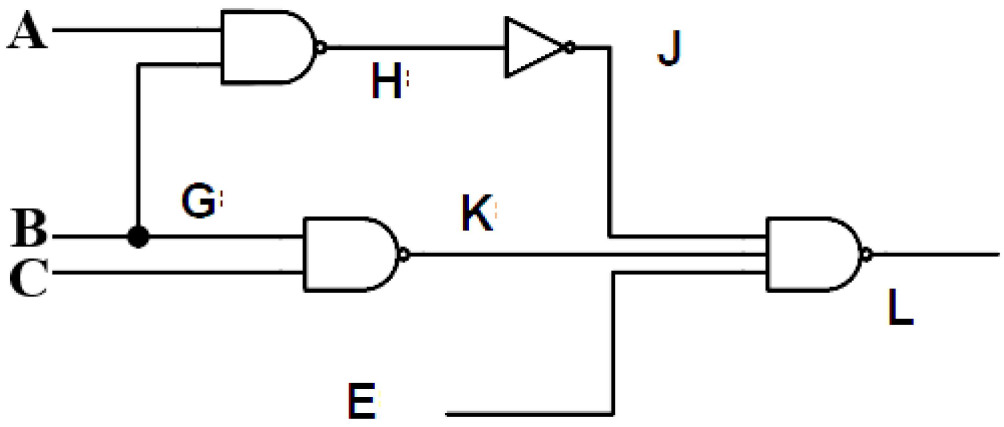
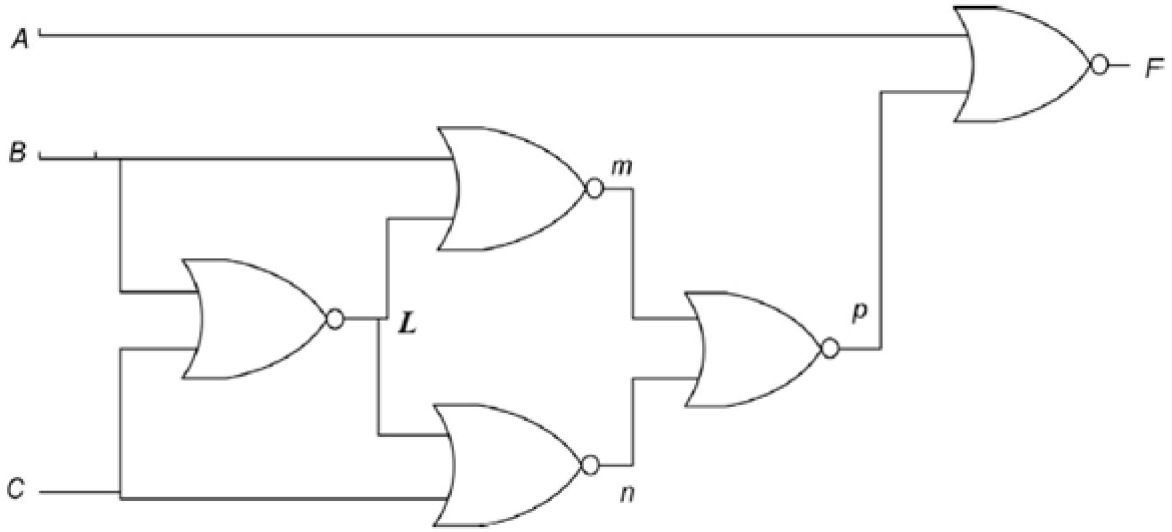


Fig. 4C.

5) For the circuit shown in Fig. 5A, where node L is stuck-at-1 (SA1), determine the test vector using the PODEM algorithm.

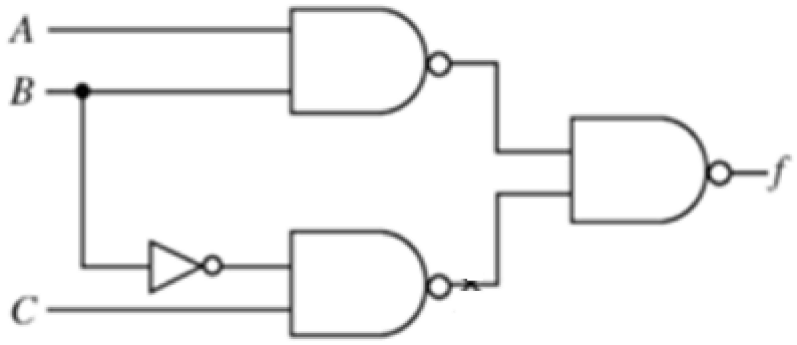
A) 

(4)

Fig. 5A.

B) Explain the any two key Design-for-Testability (DFT) guidelines used for testing digital circuits. How do these guidelines help in improving fault detection and diagnosis during the testing process. (3)

C) For the circuit shown in Fig. 5C, find the Boolean difference $\frac{df}{dA}$.



(3)

Fig. 5C.

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